SUBHRONIL MUKHOPADHYAY

AI/ML Professional



AI/ML engineer delivering **Agentic AI, Computer Vision, and LLM Systems** from research prototype to working software.

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Profile Summary

- Sr. Technical Engineer at **Hitachi India Pvt. Ltd.**, building **agentic LLM systems** (LangGraph, MCP), zero-shot **visual inspection**, and LLM serving and safety research.
- Reproduced and extended recent research, including **Multi-Agent Debate / iMAD (AAAI 2026)** and the **UNISafe robot safety filter (CoRL 2025)**, on open-source models and Isaac Sim.
- Hands-on across the stack: **PyTorch, vLLM, DINOv2, Gemini API**, FastAPI, Docker, React/Node, and deployments on **AWS and GCP**.
- Effective **communication and cross-functional collaboration**, coordinating with research, engineering, and Japan R&D teams.

Work Experience

Jul'26-Present

Sr. Technical Engineer (Contract via Hitachi Systems) | Hitachi India Pvt. Ltd., Bengaluru

Key Result Areas:

- Engineered an **agentic travel middleware** that turns one natural-language prompt into an end-to-end rail itinerary across **3 MCP servers exposing 9 tools**, via a **stateful 9-node LangGraph workflow** with guardrails and deterministic + LLM-driven routing over 9 use cases.
- Designing the **Phase-2 operator–traveler matching engine** for rail ticketing: built the hub data model and a prototype that balances traveler fit against operator goals via a **Pareto skyline and friction scoring**.
- Built a **zero-shot visual inspection system** that catches missing or wrongly fitted parts from a single golden photo with no training data; over 15 versions, cut false alarms per good frame **17× (6.9 → 0.40)** and nearly doubled defect-class **F1 (0.49 → 0.89)** using DINOv2/ResNet18 features, EfficientLoFTR + MAGSAC++ registration, and random-field-theory thresholds.
- Reproduced **Multi-Agent Debate and iMAD (AAAI 2026)** on Qwen3-8B across 6 benchmarks; iMAD cut debate tokens by **18–63%** while capturing **~90% of MAD's accuracy gain** on GSM8K, then built a 117-test hidden-state analysis pipeline.
- Owned **serving engineering and GPU experiments** for a study of whether **activation-probe safety monitors** stay calibrated in vLLM (FP8 KV cache, co-batching, torch.compile), and built a guard against silent recompilation.
- Reproduced the **UNISafe robot safety filter (CoRL 2025)** in Isaac Sim/Isaac Lab with **1,000-trial evaluations**, ruling out 5 causes of the gap to the paper and tracing failures to policy gripper-open commands.
- Pressure-tested a **GPU-to-AWS Trainium migration** for LLM inference, uncovering **4 blocking gaps** in the Neuron SDK, surveying 8+ accelerators, and proposing a scoped path at **half the effort** (2–3 vs. 3–6 person-months).

Internships

Jan'26 – Jul'26

Research and Development Intern | Hitachi India Pvt. Ltd., Bengaluru

Key Result Areas:

- Built an ETL pipeline over **227K+ ATM electronic-journal files from 27 machines** (decryption, merging, error-code extraction) and trained 8 failure-prediction models, with **LightGBM reaching 0.83 ROC-AUC**.

Areas of Exposure

- Agentic AI & LLM Systems
- Computer Vision & Inspection
- LLM Serving & Evaluation
- Reinforcement Learning
- Robotics Simulation
- Predictive Analytics
- Research Reproduction
- Full-stack Development

Soft Skills

- Analytical Thinking
- Willingness to Learn
- Innovative Thinking
- Team Player
- Time Management
- Problem-solving

- Built a **video-to-SOP pipeline for GSK equipment maintenance**: Gemini 2.5 extracts atomic actions and Q-learning process discovery with Petri-net conformance beats the heuristic baseline; rule-based checks flag missed or out-of-sequence steps.
- Produced a **1,000+ line Manim explainer** animating the 5-stage process-discovery pipeline, with 3D embeddings and an FSM resolving 10 canonical steps into 4 validated task orderings.
- Learned **procedural task graphs from egocentric videos** (CaptainCook4D) with RL, reaching **loop-F1 of 1.00** and accepting 12% of invalid traces vs. 94% for a DAG baseline.
- Programmed **real-time acoustic source localization** (4-mic array, GCC-PHAT + CNN-BiGRU) for touchless task validation, and a dashboard consolidating **10+ data-center KPIs**.

Jun'24 – Jul'24
Machine Learning Intern | BeyonData Solutions Pvt. Ltd., Ahmedabad
Highlights:

- Conducted comprehensive benchmarking of **5 CNN architectures** (VGG16, ResNet50, InceptionV3, MobileNetV2, EfficientNetB0) on a **27-class logo dataset, evaluating performance across multiple metrics.**
- Developed and optimized **data augmentation pipelines** to improve model generalization, achieving a peak classification **accuracy of 93.8% using MobileNetV2**.

Academic Projects

Almanac – Life Dashboard | Node, React, TypeScript, SQLite/Turso, GCP Cloud Run - November 2026
Highlights:

- Built and deployed a **multi-user life dashboard** (spending, trends, commitments, routines) on **Google Cloud Run**, backed by Turso-hosted SQLite with a local replica and daily snapshots to Cloud Storage.
- Wrote a **zero-dependency Node 22 server** with scrypt auth, signed sessions, and lockout, plus trend lines gated on **$r^2 \geq 0.25$** so noise is never reported as a trend.

TimeCapsule Connect | MERN Stack, PostgreSQL - July 2025
Highlights:

- Designed and developed a scalable full stack digital time-capsule platform with time- and location-based unlocking, **supporting 100,000+ unique unlock combinations over a 20-year span using KSUIDs and Google Maps API** for precise geolocation.
- Implemented real-time collaborative editing with **SlateJS and Yjs** (<100 ms latency for 10+ users) and **optimized large media uploads** (>1GB) via chunked processing, reducing upload failures **by 60%** and improving user experience.

Dynamic Crowd Simulation | GNN, Transformer, Pytorch, PPO, OpenCV, Python - March 2025
Highlights:

- Designed a hybrid trajectory-prediction model combining Graph Neural Networks and Transformer architectures, **achieving 99%+ accuracy in pedestrian movement forecasting.**
- Integrated a PPO-based reinforcement learning agent for collision-free navigation and trained the system over **20,000+ GPU timesteps**, resulting in a **+6.31 mean reward improvement and strong real-world robustness.**

Certifications

- **Gate Qualified** - Gate, Mar 2026
- **Amazon ML Summer School** - Amazon, Sept 2025
- **Microsoft Azure Ai900** - Microsoft, Aug 2024

Personal Details

Date of Birth: 19th December 2004
Languages Known: English, Hindi, Gujarati & Bengali

Technical Skills

Python, C++, Java & TypeScript

PyTorch, LangGraph, MCP & vLLM

AWS, GCP, Docker & FastAPI

ReactJS, NodeJS & Streamlit

OpenCV, DINOv2 & SAM 2

Git, Figma, OrCAD & Verilog HDL

Education

Complete (Sept'22 – May'26)	B.Tech. Computer Science, Vellore Institute of Technology | Vellore CGPA: 8.88

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